

Zero Thickness Does Not Force a Hilbert Bi-Lipschitz Embedding

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Status

This packet gives a counterexample to the second open question of Margaris–Robinson [1]. Thickness exponent zero does not force the existence of a nonlinear bi-Lipschitz embedding into a Hilbert space. Since Euclidean spaces are Hilbert spaces, it also rules out the Euclidean target.

Source Question

After proving an embedding theorem for compact subsets of Banach spaces with thickness exponent less than 1, the source paper asks:

(0)

There are a number of open questions that arise naturally from the results above. The most important are the following.

1. Can we extend Theorem 3.2 without the restriction of the thickness exponent being less than 1, in such a way that it improves on Theorem 2.4?
2. Can we prove the existence of a nonlinear bi-Lipschitz embedding into either a Euclidean or a Hilbert space when the thickness equals zero?

Figure 1: Source evidence from arXiv:1811.03872, PDF page 20.

The construction below addresses the second question.

The Counterexample

Let $Q_m = \{0, 1\}^m$ be the m -dimensional Hamming cube, equipped with the Hamming metric d_m . Let 0_m denote its zero vertex. Put

$$E_m = \ell_\infty(Q_m)$$

and embed Q_m into E_m by the pointed Kuratowski map

$$\iota_m(x) = (d_m(x, y) - d_m(0_m, y))_{y \in Q_m}.$$

This map is isometric: for every $x, x' \in Q_m$,

$$\|\iota_m(x) - \iota_m(x')\|_\infty = \sup_{y \in Q_m} |d_m(x, y) - d_m(x', y)| = d_m(x, x'),$$

where the reverse inequality is the triangle inequality and equality follows by taking $y = x$.

Let

$$B = \left(\bigoplus_{m=1}^{\infty} E_m \right)_{c_0}$$

with norm $\|(u_m)\| = \sup_m \|u_m\|_{E_m}$. Choose

$$R_m = 2^{-m^3}, \quad s_m = \frac{R_m}{m}.$$

For $x \in Q_m$, let $u_{m,x} \in B$ be the vector whose m -th coordinate is $s_m \iota_m(x)$ and whose other coordinates are zero. Define

$$X = \{0\} \cup \{u_{m,x} : m \in \mathbb{N}, x \in Q_m\}.$$

Theorem 1. *The compact set $X \subset B$ has thickness exponent 0, but X does not admit a bi-Lipschitz embedding into any Hilbert space.*

Proof. First, X is compact. Each level $\{u_{m,x} : x \in Q_m\}$ is finite, and

$$\|u_{m,x}\| \leq s_m \text{diam}(Q_m) = s_m m = R_m \rightarrow 0.$$

Thus all levels eventually lie in an arbitrarily small ball around 0.

We next compute a box-counting bound. If

$$R_{M+1} < \varepsilon \leq R_M,$$

then all levels $m > M$ are contained in the ε -ball around 0. The first M levels can be covered point by point. Hence

$$N(X, \varepsilon) \leq 1 + \sum_{m=1}^M |Q_m| = 1 + \sum_{m=1}^M 2^m \leq 2^{M+1}.$$

Since $-\log R_M = M^3 \log 2$, it follows that

$$\overline{\dim}_B(X) = \limsup_{\varepsilon \downarrow 0} \frac{\log N(X, \varepsilon)}{-\log \varepsilon} \leq \limsup_{M \rightarrow \infty} \frac{(M+1) \log 2}{M^3 \log 2} = 0.$$

The thickness exponent is always bounded above by the upper box-counting dimension: an ε -cover by $N(X, \varepsilon)$ balls gives an ε -approximating subspace spanned by the $N(X, \varepsilon)$ centres. Therefore $\tau(X, B) = 0$.

It remains to exclude Hilbert embeddings. Suppose, towards a contradiction, that $F : X \rightarrow H$ is bi-Lipschitz into a Hilbert space H . Then there are constants $0 < a \leq b < \infty$ such that

$$a\|u - v\|_B \leq \|F(u) - F(v)\|_H \leq b\|u - v\|_B \quad (u, v \in X).$$

Restrict F to the m -th level. Since ι_m is isometric, for all $x, x' \in Q_m$,

$$as_md_m(x, x') \leq \|F(u_{m,x}) - F(u_{m,x'})\|_H \leq bs_md_m(x, x').$$

After dividing by s_m , this gives a Hilbert embedding of Q_m with distortion at most b/a , uniformly in m .

The following standard Enflo cube estimate contradicts this. If $G : Q_m \rightarrow H$ has Lipschitz constant at most β and inverse Lipschitz constant at least α , then $\beta/\alpha \geq \sqrt{m}$. Indeed, identify Q_m with $\{-1, 1\}^m$. For $\varepsilon \in Q_m$, let $-\varepsilon$ be the antipodal vertex and $\varepsilon^{(j)}$ the vertex obtained by flipping the j -th coordinate. Hilbert-valued Walsh expansion gives

$$\mathbb{E}_\varepsilon \|G(\varepsilon) - G(-\varepsilon)\|^2 \leq \sum_{j=1}^m \mathbb{E}_\varepsilon \|G(\varepsilon) - G(\varepsilon^{(j)})\|^2.$$

The right side is at most $m\beta^2$, since each edge has Hamming length 1. Therefore some antipodal pair satisfies

$$\|G(\varepsilon) - G(-\varepsilon)\| \leq \beta\sqrt{m}.$$

But the lower Lipschitz bound gives

$$\|G(\varepsilon) - G(-\varepsilon)\| \geq \alpha m.$$

Thus $\beta/\alpha \geq \sqrt{m}$.

Applying this to the uniformly distorted embeddings of the cubes Q_m forces $b/a \geq \sqrt{m}$ for all m , impossible. Hence no such Hilbert embedding F exists. $\square$

Verification Notes

The construction deliberately separates global and local geometry. The levels shrink quickly enough that the global upper box-counting dimension is zero, but each level retains the full Hamming-cube metric after rescaling. Thickness sees the global covering rate; Hilbert bi-Lipschitz embeddability would have to handle all finite cube levels with one distortion constant.

Novelty and Search Bounds

Before packaging, the run's cheap indexes were searched for arXiv:1811.03872, “zero thickness”, “bi-Lipschitz”, “Euclidean”, “Hilbert”, and “Assouad”. No prior packet or attempt for this target was found. The proof uses the standard Enflo lower bound for Hamming cubes in Hilbert space and is packaged as an original counterexample to the source question.

Human Review Recommendation

Review as a counterexample packet. The essential checks are the box-counting/thickness computation and the uniform-distortion obstruction from the Hamming cube levels.

References

- [1] A. Margaris and J. C. Robinson, *Embedding properties of sets with finite box-counting dimension*, arXiv:1811.03872.